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Lab Report: RF Transmission, Reflections, and Standing Waves

Experiment performed: Part A: Phase Velocity and Velocity Factor, Part B: Reflection Coefficient and Characteristic Impedance, Part C: Inductance and Capacitance per Unit Length and Part D: Cable Attenuation

1. Objective

The objective of this experiment is to study signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies and to determine the following cable parameters experimentally:

- To determine the phase velocity and velocity factor of the coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.
- To study signal reflection from an open-ended cable and from a cable terminated with a 50 Ω load, and to estimate the magnitude of the reflection coefficient in each case.
- To verify that the coaxial cable is approximately matched to a 50 Ω termination, indicating that its characteristic impedance is close to 50 Ω.
- To calculate the inductance and capacitance per unit length of the cable using the measured phase velocity and the approximate value $Z_0 \approx 50\ \Omega$.
- To measure the insertion loss and attenuation per unit length of the coaxial cable at different frequencies.

2. Theory

2.1 Transmission Lines and Reflections

A coaxial cable supports transverse electromagnetic (TEM) propagation, in which both the electric and magnetic fields are transverse to the direction of propagation. The signal velocity in the cable is lower than the speed of light in vacuum because of the dielectric material between the conductors. When a travelling wave meets a load impedance Z_L different from the characteristic impedance Z_0 of the cable, part of the wave reflects back toward the source. At a fixed observation point (the Tee junction, in this experiment), the incident and reflected waves superimpose to form a standing-wave pattern. As the frequency is swept, the phase of the reflected wave changes, so the amplitude measured at the Tee junction varies periodically between constructive and destructive interference.

2.2 Phase Velocity Measurement

For a cable of physical length ℓ and phase velocity v_p, left open at the far end, the round-trip time delay is $\tau = 2\ell/v_p$, and the round-trip phase of the reflected signal is $\phi_{RT} = 4\pi f\ell/v_p$ (taking the open-circuit phase shift $\phi_\Gamma = 0$). Two consecutive maxima, or two consecutive minima, are separated by a round-trip phase change of 2π, which gives:

$$\tau = 1/\Delta f, \qquad v_p = 2\ell\Delta f, \qquad VF = v_p/c$$

where Δf is the frequency spacing between consecutive maxima or minima, VF is the velocity factor, and $c \approx 3 \times 10^8$ m/s is the speed of light in vacuum.

2.3 Reflection Coefficient and Characteristic Impedance

The voltage reflection coefficient at the load is $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$. For an open circuit, $\Gamma = +1$; for a short circuit, $\Gamma = -1$; and for a matched load ($Z_L = Z$), $\Gamma = 0$. During a frequency sweep, the oscilloscope displays Vmax = |Vincident| + |Vreflected| at constructive interference and Vmin = |Vincident| − |Vreflected| at destructive interference, so the magnitude of the reflection coefficient observed at the Tee junction is:

$$|\Gamma| = (Vmax - Vmin) / (Vmax + Vmin)$$

A significant reduction in $|\Gamma|$ when the cable is terminated in 50 Ω, compared to the open-circuit case, verifies that the cable is approximately matched to 50 Ω and that its characteristic impedance Z_0 is close to 50 Ω.

2.4 Inductance and Capacitance per Unit Length

For a low-loss line, $vp = 1/\sqrt{(L'C')}$ and $Z_0 = \sqrt{(L'/C')}$, where L' and C' are the inductance and capacitance per unit length. Using the measured vp and the approximate $Z_0 \approx 50$ Ω:

$$L' = Z0 / v_p, \qquad C' = 1 / (Z_0\, v_p)$$

2.5 Cable Attenuation

Cable attenuation is the reduction in signal power as the wave propagates through the cable, measured here in dBm with a spectrum analyser. The insertion loss is IL = Pref − Pcable, where Pref is the power measured through a short reference connection and Pcable is the power measured after inserting the cable under test. The attenuation per unit length is:

$$\alpha = IL / \ell \text{ (dB/m)}$$

3. Components Required

- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser
- RF Tee connector
- Coaxial cable under test (RG 174)

- Short reference coaxial cable
- 50 Ω termination
- Open-circuit termination (cable end left unconnected)
- Suitable RF adapters and connectors

4. Experimental Setup

The signal generator, cable under test, and oscilloscope were connected through the Tee connector, with the far end of the cable terminated according to the measurement being performed (open, 50 Ω, or short). The oscilloscope input impedance was set to 1 MΩ throughout so as not to load the Tee junction and alter the reflection pattern. The signal-generator amplitude was held constant during each frequency sweep.

Because the DSO used has a bandwidth limit of 200 MHz, the cable length was chosen to keep all standing-wave maxima and minima of interest well within this range (see Learnings). A cable of length 305 cm was used for the phase-velocity and reflection-coefficient measurements (Parts A and B); a separate, shorter cable of length 91.5 cm was used for the attenuation measurement (Part D).

5. Experimental Procedure

5.1 Part A - Phase Velocity and Velocity Factor

With the far end of the cable (L = 305 cm) left open and the oscilloscope input impedance set to 1 MΩ, a constant-amplitude sinusoidal signal was applied, and its frequency was swept from 10 MHz to 60 MHz. The Vrms amplitude at the Tee junction was recorded at 1 MHz intervals, and the frequencies of consecutive maxima/minima were identified from the resulting sweep.

5.2 Part B - Reflection Coefficient

Using the same Tee-junction setup, Vmax and Vmin were recorded over one complete ripple, first with the cable left open and then with a 50 Ω termination connected at the far end (signal-generator amplitude unchanged). $|\Gamma_{open}|$ and $|\Gamma_{50}|$ were then calculated and compared.

5.3 Part C - Inductance and Capacitance per Unit Length

Using the measured v_p from Part A and the approximate value $Z_0 \approx 50\ \Omega$ established from the matching test in Part B, L' and C' were calculated.

5.4 Part D - Cable Attenuation

The signal generator was connected to the spectrum analyser through a short reference cable, and the reference power Pref was recorded at each test frequency with the generator

power fixed at 0 dBm. The short reference connection was then replaced with the cable under test (ℓ = 91.5 cm), and the power Pcable was recorded at the same frequencies (100, 150, 200, 250, 300 MHz), keeping the rest of the setup unchanged.

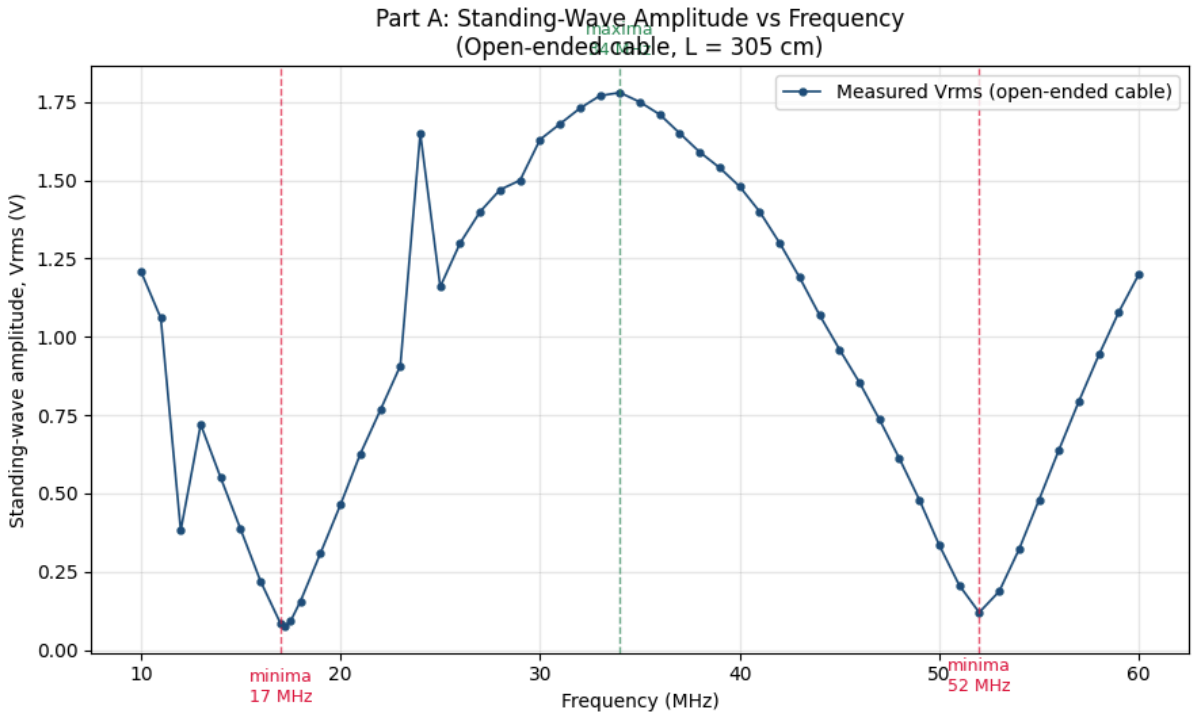
6. Observations and Graphs

6.1 Part A - Amplitude vs Frequency (Open-ended cable, L = 305 cm)

Frequency (MHz)	Vrms
10	1.21 V
11	1.06 V
12	381.60 mV
13	720.45 mV
14	550.41 mV
15	385.60 mV
16	219.36 mV
17 (minima)	84.7 mV
17.2	76.08 mV
17.5	91.90 mV
18	156.07 mV
19	309.07 mV
20	463.67 mV
21	626.76 mV
22	766.22 mV
23	907.45 mV
24	1.65 V
25	1.16 V
26	1.30 V
27	1.40 V
28	1.47 V
29	1.50 V

30	1.63 V
31	1.68 V
32	1.73 V
33	1.77 V
34 (maxima)	1.78 V
35	1.75 V
36	1.71 V
37	1.65 V
38	1.59 V
39	1.54 V
40	1.48 V
41	1.40 V
42	1.30 V
43	1.19 V
44	1.07 V
45	960.1 mV
46	854.5 mV
47	736.6 mV
48	612 mV
49	480 mV
50	335 mV
51	207 mV
52 (minima)	120 mV
53	187 mV
54	322 mV
55	480 mV
56	640 mV
57	795 mV
58	943 mV
59	1.08 V

60	1.20 V



6.2 Part B - Reflection Coefficient

Termination	Vmax	Vmin	|Γ|
Open circuit	1.72 V	45.3 mV	0.9487
50 Ω load	1.06 V	933.36 mV	0.0635

6.3 Part D - Cable Attenuation (ℓ = 91.5 cm)

Generator output power set to 0 dBm; a fixed loss of 1.4 dB was measured through the connecting wire itself, giving Pref = −1.4 dBm at every test frequency.

Frequency (MHz)	Pref (dBm)	Pcable (dBm)
100	-1.4	-1.65
150	-1.4	-1.8
200	-1.4	-1.96
250	-1.4	-2.12
300	-1.4	-2.18

7. Calculations and Results

7.1 Part A - Phase Velocity and Velocity Factor

Minima were identified at 17 MHz and 52 MHz, and a maxima at 34 MHz, consistent with the maxima lying midway between the two minima (34 − 17 = 17 MHz and 52 − 34 = 18 MHz, averaging to a half-period of 17.5 MHz). Using the two consecutive minima directly:

$$\Delta f = 52\ \text{MHz} - 17\ \text{MHz} = 35\ \text{MHz} = 3.5 \times 10^{7}\ \text{Hz}$$

With cable length $\ell = 305$ cm $= 3.05$ m:

$$v_p = 2\ell\Delta f = 2 \times 3.05 \times 3.5 \times 10^{7} = 2.135 \times 10^{8}\ \text{m/s}$$

$$VF = v_p / c = 2.135 \times 10^{8} / 3 \times 10^{8} \approx 0.712$$

This measured velocity factor (≈ 0.71) is in reasonable agreement with the manufacturer-specified velocity factor of 0.66 for RG 174 (see Reference 1), the ~8% difference being attributable to the 1 MHz frequency-sweep resolution used to locate the extrema and to the non-ideal open-circuit termination discussed in the Learnings section. This agreement supports the identification of the cable under test as RG 174.

7.2 Part B - Reflection Coefficient

Open circuit:

$$|\Gamma_{open}| = (1.72 - 0.0453) / (1.72 + 0.0453) = 1.6747 / 1.7653 \approx 0.9487$$

50 Ω load:

$$|\Gamma_{50}| = (1.06 - 0.93336) / (1.06 + 0.93336) = 0.12664 / 1.99336 \approx 0.0635$$

The reflection coefficient falls from ≈ 0.95 (open circuit, close to the ideal $\Gamma = 1$ for total reflection) to ≈ 0.06 (50 Ω load, close to the ideal $\Gamma = 0$ for a matched load). This large drop confirms that the coaxial cable is approximately matched to a 50 Ω termination, i.e., its characteristic impedance Z_0 is close to 50 Ω.

7.3 Part C - Inductance and Capacitance per Unit Length

Using $v_p = 2.135 \times 10^{8}$ m/s from Part A and $Z_0 \approx 50\ \Omega$ from Part B:

$$L' = Z_0 / v_p = 50 / (2.135 \times 10^{8}) \approx 2.342 \times 10^{-7}\ \text{H/m} = 234.2\ \text{nH/m}$$

$$C' = 1 / (Z_0\ v_p) = 1 / (50 \times 2.135 \times 10^{8}) \approx 9.368 \times 10^{-11}\ \text{F/m} = 93.68\ \text{pF/m}$$

7.4 Part D - Cable Attenuation

Insertion loss IL = Pref − Pcable, and attenuation per unit length $\alpha = IL / \ell$, with $\ell = 0.915$ m:

Frequency (MHz)	IL (dB)	α (dB/m)
100	0.25	0.273
150	0.40	0.437
200	0.56	0.612
250	0.72	0.787
300	0.78	0.852

The attenuation per unit length increases steadily with frequency, from about 0.27 dB/m at 100 MHz to about 0.85 dB/m at 300 MHz, consistent with the expected increase in coaxial-cable loss with frequency due to the skin effect and dielectric loss.

8. Results Table

Quantity	Symbol	Value
Cable length (Part A/B)	ℓ	3.05 m (305 cm)
Frequency spacing	Δf	3.5×10^7 Hz (35 MHz)
Phase velocity	v_p	2.135×10^8 m/s
Velocity factor	$VF = v_p/c$	≈ 0.712
Reflection coefficient (open)	$\lvert\Gamma_{open}\rvert$	≈ 0.9487
Reflection coefficient (50 Ω)	$\lvert\Gamma_{50}\rvert$	≈ 0.0635
Approximate characteristic impedance	Z_0	≈ 50 Ω
Inductance per unit length	L'	≈ 234.2 nH/m
Capacitance per unit length	C'	≈ 93.68 pF/m
Attenuation (100–300 MHz)	α	0.27 – 0.85 dB/m

9. Conclusion

- The standing-wave amplitude at the Tee junction, measured while sweeping the drive frequency into an open-ended cable, showed clear periodic maxima and minima due to interference between the incident and reflected waves (minima at 17 MHz and 52 MHz, maxima at 34 MHz).
- The measured phase velocity was $v_p \approx 2.135 \times 10^8$ m/s, giving a velocity factor VF ≈ 0.712, in reasonable agreement with the RG 174 datasheet value of 0.66.
- The reflection coefficient dropped from ≈ 0.95 (open circuit) to ≈ 0.06 (50 Ω load), verifying that the cable under test is approximately matched to a 50 Ω termination and that its characteristic impedance Z_0 is close to 50 Ω.

- The calculated inductance and capacitance per unit length were $L' \approx 234.2$ nH/m and $C' \approx 93.68$ pF/m.
- The measured attenuation per unit length increased from ≈ 0.27 dB/m at 100 MHz to ≈ 0.85 dB/m at 300 MHz, consistent with the expected frequency dependence of coaxial-cable loss.

10. Learnings

We validated our RG 174 cable by measuring its velocity factor, which came out close to the manufacturer-specified value. I also learnt that leaving the end of the Tee connector open gives a phase shift of zero at the very end point, whereas terminating it with a 50 Ω resistor introduces a 180° phase shift instead.

For Part A we applied V = 1 V, so ideally the maxima should reach 2 V and the minima should reach 0 V from perfect constructive and destructive addition. In practice the measured maxima and minima differ from these ideal values, because the wave changes somewhat as it travels down the cable (attenuation), so the incident and reflected waves never add up in a perfectly ideal constructive or destructive way.

Since the DSO used has a bandwidth limit of only 200 MHz, we deliberately increased the cable length: a longer cable produces more maxima and minima within a given frequency range, so the same standing-wave features could be observed at lower, in-range frequencies. This is why we ultimately used a 305 cm cable for Part A/B rather than our earlier 95 cm test cable; the shorter cable pushed the relevant maxima and minima above 200 MHz, beyond the DSO's range, and gave falsified/unreliable results.

We also tried a short-circuit load at the Tee connector. Since a short load has 0 Ω resistance, we found that even without an added 50 Ω resistance, a 180° phase shift is still obtained on reflection, though some reflection (rather than a clean match) is still present in that case.

11. References

1. Faber Kabel, RG 174 Datasheet (velocity factor 0.66): https://shop.faberkabel.de/out/downloads/uploads/EN/dbl_rg174.pdf